Reg No……..

Name……..

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL QUESTION PAPER

Fourth Semester

Discipline Specific Core Course (DSC) - MG4DSCECO201 – QUANTITATIVE ECONOMICS - II

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum marks: 70

(Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I) Appreciation (Ap) and Skill (S))

Students should attempt at least one question from each course outcome to enhance their overall outcome availability.

[Learning Domain] [Cos No (s)]

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. Which of the following is an example of primary data? [Ap][2]
   a) Census report
   b) Researcher's field survey
   c) Government publication
   d) World Bank database
2. Which of the following is NOT a type of index number? [U][1, 4]
   a) Price index
   b) Quantity index
   c) Value index
   d) Cost index

3. A time series is a set of observations: [Ap][2]
   a) Arranged according to magnitude
   b) Collected at one point of time
   c) Arranged in chronological order
   d) Classified according to attributes
4. For a sample of 10 observations, degrees of freedom in a t-test is [U][1]
   a) 10
   b) 9
   c) 8
   d) 11
5. The value of Karl Pearson's correlation coefficient lies between: [U][1,4]
   a) 0 and 1
   b) –1 and +1
   c) $-\infty$ and $+\infty$
   d) 0 and –1
6. Skewness refers to: [U][1]
   a) Degree of dispersion
   b) Symmetry or asymmetry of a distribution
   c) Height of the frequency curve
   d) Variability of data
7. The regression coefficient of Y on X indicates: [A][3]
   a) Degree of association between X and Y
   b) Change in X due to Y
   c) Change in Y due to a unit change in X
   d) Total variation in Y
8. A continuous random variable can take: [U][1]
   a) Only whole numbers
   b) Only integers
   c) Countable values
   d) Any value within a given range
9. Kurtosis is useful in: [U][1]
   a) Measuring dispersion
   b) Studying concentration of data around mean

c) Finding correlation

d) Measuring skewness

10. If r = –0.85, the correlation is: [U][1,4]

a) Weak positive

b) Strong positive

c) Strong negative

d) Zero (10*1=10)

Part B

Short Answer Questions (50 words)

Answer any ten questions

Each question carries 3 marks

11. Find the AM of first ten natural numbers. [U][1]
12. Define ogive and distinguish between less-than and greater-than ogives. [Ap][2]
13. Explain the properties of binomial distribution. [Ap][2]
14. State any two uses of correlation. [U][1,4]
15. Define Probability Density Function and state its properties. [Ap][2]
16. Compare random and non-random sampling methods and examine their suitability in socio-economic research. [Ap][2]
17. From the following data, find Quartile Deviation. [Ap][2]

    12, 18, 25, 30, 35, 42, 48, 50
18. Write the formula for Karl Pearson's correlation coefficient. [U][1,4]
19. Estimate Median [U][1]

No of Families	0	1	2	3	4	5	6
No of Children	3	8	12	9	4	2	1

20. From the following data, calculate the Quartile Deviation. [Ap][2]

Class	0-10	10-20	20-30	30-40
Frequency	5	9	14	7

21. Explain the properties of t-distribution. [Ap][2]
22. From the following data, calculate the Coefficient of Variation and state which series is more consistent:

    Series A : Mean = 40, SD = 8

Series B : Mean = 50, SD = 12 [Ap][2]

23. Draw a histogram from the following table. [Ap][2]

Class	0-20	20-40	40-60	60-80	80-100
Freq	5	20	32	16	4

(10*3=30)

Part C

Short Essay Type Questions (160 words)

Answer any 5 questions

Each question carries 6 marks

24. Estimate Standard Deviation [U][1]

Class	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89
Freq	4	2	18	20	22	16	15	14	11

25. Draw a trend line using the method of semi averages from the following data. [Ap][2]

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005
Sale	100	105	109	96	102	108	112	114	116

26. The mean marks obtained by students in an examination are 60 and the standard deviation is 8. Find the probability that a student scores more than 68 marks. [A[3]

27. Given the following data:

X	5	7	9	11	13
Y	15	18	21	24	27

a. Find the regression equation of Y on X
b. Estimate the value of Y when X = 15
c. Interpret the regression coefficient. [A][3]

28. Data showing the annual production of a first for 10 years given below. Draw a trend line. [Ap][2]

Years	1995	1996	1997	998	1999	2000	2001	2002	2003	2004
Production	14	22	20	26	18	30	21	32	28	35

29. From the following pairs of observations, state whether the correlation is positive, negative, or zero**:** [An][1,4]

X	5	10	15	20	25
Y	50	45	40	35	30

30. Construct index number of prices from the following data using a) Laspeyre's method b) Paasche's method c) Fisher's method and d) Marshall Edgeworth method. [U][1, 4]

Commodity	2010		2015	
	Price	Quantity	Price	Quantity
A	5	8	6	6
B	8	15	10	8
C	4	15	5	10
D	2	10	4	13

(6*5=30)